

Question 1: Data Preparation and Overview

The dataset consists of 442 diabetic patients measured at baseline and evaluated for disease progression one year later. Each row represents a single patient. The dependent variable is 'progression' (originally 'y'), a quantitative measure of disease progression. Predictor variables include demographic factors (age, sex), clinical measurements (BMI, blood pressure), and six serum measurements (s1 to s6). While 'sex' was numerically coded (1 and 2), it represents a categorical factor and was converted accordingly. An initial audit confirmed zero missing values (NA = 0) and zero duplicated entries. Because health data carries sensitive clinical implications, cautious interpretation is required: predictive associations must not be mistaken for clinical causation.

- **What does one row represent?**

Each row in the dataset represents a single individual: one of the 442 diabetic patients evaluated in the study.

- **Which variable is the outcome?**

The outcome (dependent variable) is progression (originally named y in the raw dataset). It is a quantitative, continuous measurement indicating the progression of diabetes one year after baseline measurements were taken.

- **Which variables are predictors?**

There are 10 baseline predictor (independent) variables:

Demographic: age (in years), sex (biological sex).

Clinical: bmi (body mass index in kg/m^2), bp (mean arterial blood pressure).

Blood serum measurements: s1 (total serum cholesterol), s2 (low-density lipoproteins / LDL), s3 (high-density lipoproteins / HDL), s4 (total cholesterol / HDL ratio), s5 (log of serum triglycerides level), and s6 (blood glucose level).

- **Which variables are numerical, and which are categorical in meaning?**

Categorical variable: sex. Although it is coded numerically in the raw file (values 1 and 2), it represents distinct demographic categories rather than a continuous scale. Therefore, it was converted into a categorical factor in R (Female and Male).

Numerical continuous variables: All remaining 9 predictors (age, bmi, bp, s1, s2, s3, s4, s5, s6) and the outcome (progression) are numerical measurements.

- **Are there missing values, duplicated rows or obviously problematic values?**

An audit of data integrity showed that the dataset contains zero missing values (NA = 0) across all columns and zero duplicated rows (duplicated = 0). All clinical values fall within biologically plausible human ranges (e.g., minimum BMI is 18.0 and maximum is 42.2). The dataset is clean and ready for analysis.

- **Why do health-related data require careful and responsible interpretation?**

Health data directly influences clinical decisions, patient care, and healthcare policy. Because observational health data is subject to unmeasured confounding and selection bias, mistaken interpretations—especially confusing statistical correlation with medical causation—can lead to harmful or ineffective clinical guidelines. Furthermore, patient privacy, ethical stewardship, and transparent reporting of statistical uncertainty are essential.

Question 2: Explore the Data (EDA)

Exploratory data analysis indicates that the outcome variable, diabetes progression, exhibits substantial variability (Mean = 152.1, SD = 77.09, range 25.0 to 346.0) with a moderate positive skew, where the median (140.5) lies below the mean. BMI averages 26.38 kg/m² with a standard deviation of 4.42 kg/m², placing the typical patient in the overweight category (BMI > 25 kg/m²). The distribution of BMI is approximately bell-shaped with slight right-tail elongation up to 42.2 kg/m², highlighting meaningful clinical variation across the patient cohort.

Progression: Mean = 152.13, Median = 140.50, SD = 77.09, IQR = 124.50, Min = 25, Max = 346

BMI: Mean = 26.38, Median = 25.70, SD = 4.42, IQR = 6.07, Min = 18.0, Max = 42.2

Age: Mean = 48.52, Median = 50.00, SD = 13.11, IQR = 20.75, Min = 19, Max = 79

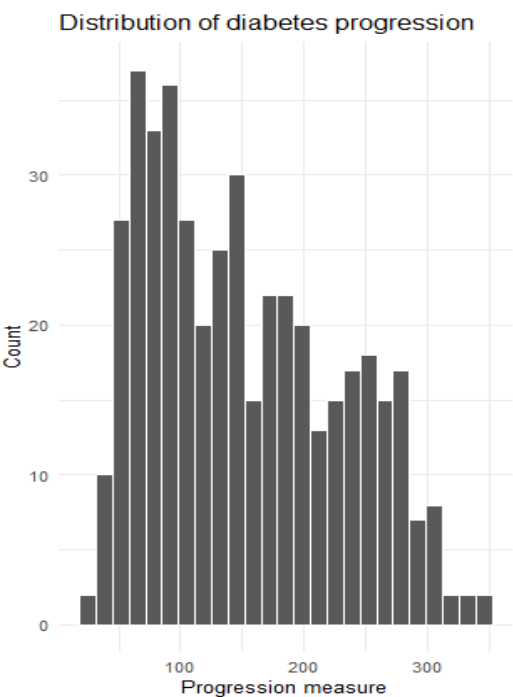


Figure 1: Progression Histogram

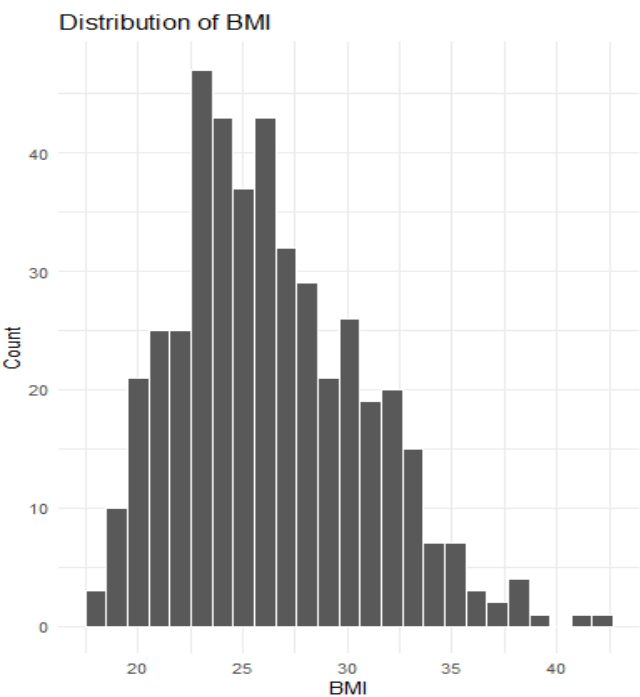


Figure 2: BMI Histogram

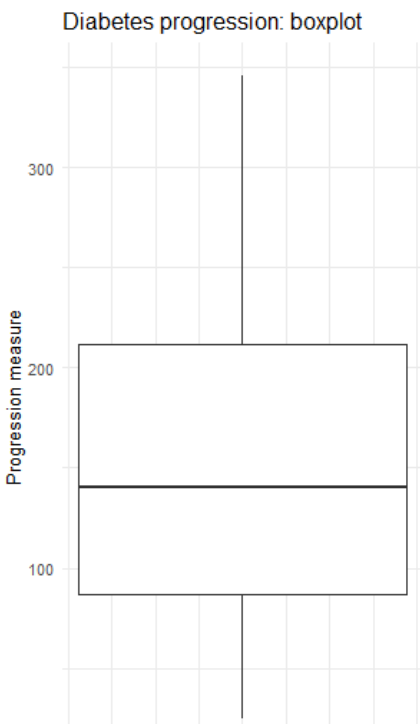


Figure 3: Progression Boxplot

- **What do the centre and spread of the main variables look like?**

* Diabetes Progression (Outcome): As displayed in Figure 1: Progression Histogram and Figure 3: Progression Boxplot, the progression measure is centered at a mean of 152.13 and a median of 140.50. The data exhibits wide dispersion across patients, with a standard deviation (SD) of 77.09 and an interquartile range (IQR) of 124.50 (spanning between 87.00 and 211.50), covering a total range from 25.00 to 346.00.

* BMI: As shown in Figure 2: BMI Histogram, body mass index has a mean of 26.38 kg/m² (SD = 4.42) and a median of 25.70 kg/m² (IQR = 6.07), ranging from 18.00 to 42.20 kg/m². This reflects a clinically realistic cohort spanning normal-weight, overweight, and obese individuals.

* Age: Patient age is centered at a mean of 48.52 years (SD = 13.11) with a median of 50.00 years (IQR = 20.75), covering an adult age range from 19.00 to 79.00 years.

- **Are the distributions approximately symmetric or clearly skewed?**

* Diabetes Progression: The distribution is moderately right-skewed (positively skewed). As visible in Figure 1: Progression Histogram, patient frequency peaks between 50 and 150, while a long right tail extends past 300. This asymmetry is further confirmed in Figure 3: Progression Boxplot, where the mean (152.13) sits visibly above the median (140.50) and the upper whisker is noticeably longer than the lower whisker.

* BMI: As seen in Figure 2: BMI Histogram, BMI is unimodal and slightly right-skewed, with most patients clustering between 22 and 28 kg/m² and a mild taper extending toward higher obesity levels.

* Age: The distribution of age is approximately symmetric and evenly spread across middle-aged adulthood.

- **Are there unusual observations?**

* In Figure 3: Progression Boxplot, there are no individual data points plotted beyond the whiskers ($1.5 \times \text{IQR}$), indicating that the target outcome contains no extreme mathematical outliers.

* In Figure 2: BMI Histogram, isolated values above 38 kg/m² represent patients with severe obesity. These reflect authentic clinical diversity rather than data entry errors or equipment malfunction.

* All 442 records represent valid, biologically plausible measurements, and no data points were removed or trimmed.

- **Which features of the data matter before modelling begins?**

1. Substantial Outcome Variance: The progression measure varies widely across a continuous scale (25 to 346) without artificial floor, ceiling, or boundary truncation, making linear regression an appropriate modelling framework.

2. Distributional Shape: The moderate right-skewness of the outcome indicates that standard linear regression can be applied, but residual diagnostics (assessing normality and constant variance of residuals) will be essential after fitting the models.

3. Data Completeness: The absence of missing values (**N = 442** complete rows) ensures that subsequent bivariate and multiple regression analyses maintain consistent sample size and full statistical power.

Question 3: Investigate Relationships

Following the univariate exploratory analysis, bivariate relationships were evaluated to identify which clinical and biochemical markers are most strongly associated with one-year diabetes progression, and to examine inter-predictor correlations that could cause multicollinearity. Pearson correlation coefficients were computed across all numeric variables, the full correlation structure was visualised in a heatmap (**Figure 4**), and key relationships were examined using scatterplots with fitted linear trend lines (**Figure 5** and **Figure 6**). The analysis indicates strong linear signals for specific metabolic and lipid markers, while highlighting substantial informational overlap among laboratory blood measurements.

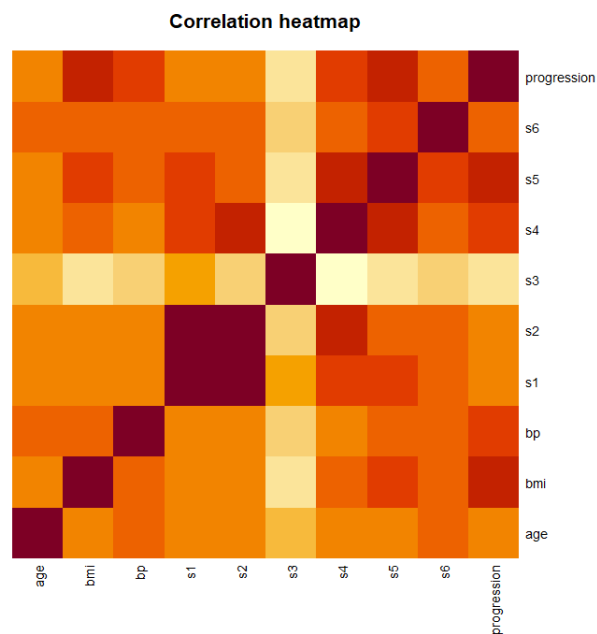


Figure 4: Correlation Heatmap of Numeric Variables

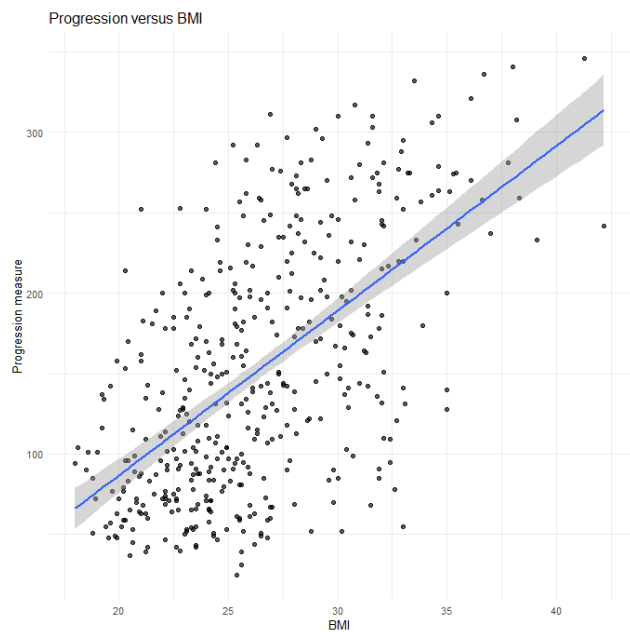


Figure 5: Progression versus BMI with Linear Trend

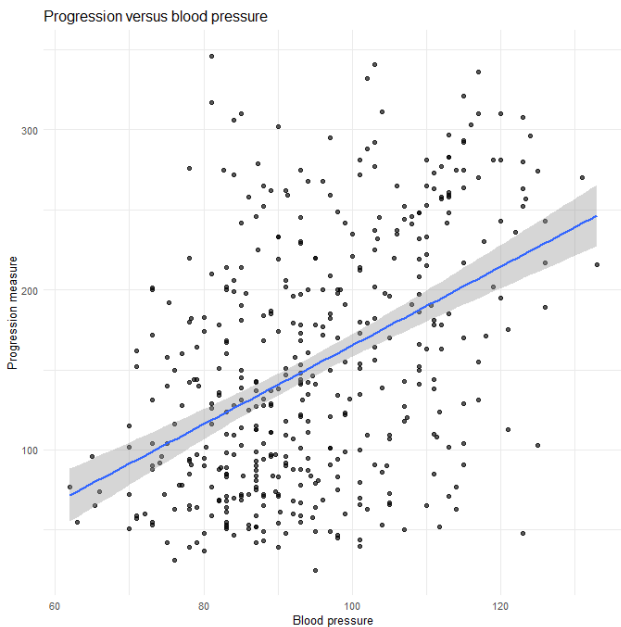


Figure 6: Progression versus Blood Pressure with Linear Trend

- **Which predictors are most strongly associated with progression?**

Based on the Pearson correlation analysis, the two strongest individual predictors of diabetes progression are **BMI** ($r = 0.59$) and serum marker **S5** ($r = 0.57$). Both exhibit substantial positive linear associations, indicating that higher body mass index and elevated S5 lipid levels correspond to markedly accelerated disease progression. Blood pressure (**BP**, $r = 0.44$), **S4** ($r = 0.43$), and **S6** ($r = 0.38$) also show moderate positive relationships, while **S3** displays a notable negative association ($r = -0.40$). Conversely, baseline age ($r = 0.19$), S1 ($r = 0.21$), and S2 ($r = 0.17$) display weak linear associations with disease progression.

- **Which predictors are strongly related to each other?**

As visualised in the correlation heatmap (**Figure 4**), inspection of the inter-predictor correlation matrix reveals significant collinearity among the serum measurements:

S1 and S2 share an exceptionally strong positive correlation ($r = 0.90$).

S3 and S4 show a strong inverse correlation ($r = -0.74$).

S4 and S5 ($r = 0.62$), **S2 and S4** ($r = 0.66$), and **S1 and S4** ($r = 0.54$) also demonstrate substantial shared variance. In contrast, clinical measurements such as age, sex, BMI, and blood pressure show modest correlations with one another ($r < 0.35$).

- **Could some predictors carry overlapping information?**

Yes. The serum lipid sub-fractions (S1 through S6) represent interrelated components of the patient's cholesterol and lipid profile. For instance, total cholesterol (S1) and LDL (S2) are functionally linked ($r = 0.90$). Including highly collinear variables simultaneously in multiple regression risks substantial multicollinearity, which can inflate standard errors and destabilize coefficient estimates.

- **Which variables look most promising for regression analysis, and why?**

BMI: It is the single strongest clinical predictor ($r = 0.59$), demonstrates a clear positive linear trajectory in **Figure 5**, and is easily measurable in routine healthcare.

S5: It provides the strongest biochemical signal ($r = 0.57$) and carries explanatory power distinct from simple anthropometric measurements.

BP (Blood Pressure): As illustrated in **Figure 6**, BP exhibits a clear moderate association ($r = 0.44$) and represents an established cardiovascular risk factor independent of lipid panels.

Sex and Age: Although age exhibits a weaker linear correlation ($r = 0.19$), both age and biological sex are essential clinical demographic controls in medical modelling.

Question 4: Build and Interpret Regression Models

To quantify the predictive capacity of clinical and biochemical markers, two ordinary least squares (OLS) linear regression models were fitted. First, a simple bivariate model was evaluated using body mass index (BMI) as the primary clinical predictor. Subsequently, a multiple linear regression model was estimated incorporating BMI, mean arterial blood pressure (BP), and serum marker S5, representing the strongest explanatory dimensions identified during exploratory analysis.

Part A: Simple Linear Regression (Model 1)

Write the fitted equation:

$$\text{Progression} = -117.77 + 10.23 * \text{BMI}$$

Interpret the slope in context:

The estimated slope ($\beta_1 = 10.23$, $p < 0.001$) indicates that for each 1 kg/m^2 increase in BMI, average one-year diabetes progression is predicted to increase by approximately 10.23 units. This confirms a statistically significant, positive clinical association between patient body mass and accelerated disease progression.

Does the intercept have a useful real-world interpretation?

The intercept is -117.77 ($p < 0.001$). In a real-world clinical context, this intercept has no meaningful practical interpretation because a BMI of 0 kg/m^2 is physiologically impossible in living humans, and the disease progression score cannot be negative. Its role is purely mathematical, anchoring the regression line across the observed clinical range of BMI (18.00 to 42.20 kg/m^2).

What does R-squared mean here?

The coefficient of determination is $R^2 = 0.3439$ (34.39%). This means that approximately 34.4% of the total variability in one-year diabetes progression is explained by differences in patient BMI alone, leaving 65.6% of the variance unexplained.

Part B: Multiple Linear Regression (Model 2)

Write the fitted equation:

$$\text{Progression} = -334.88 + 6.50 * \text{BMI} + 0.90 * \text{BP} + 49.58 * \text{S5}$$

Interpret at least two coefficients from the multiple model (using 'holding the other predictors constant'):

BMI Coefficient (6.50, $p < 0.001$): Holding blood pressure (BP) and serum marker S5 constant, each additional 1 kg/m^2 increase in BMI is associated with an expected increase of 6.50 units in one-year diabetes progression.

S5 Coefficient (49.58, $p < 0.001$): Holding BMI and blood pressure constant, each 1-unit increase in the serum triglycerides marker (S5) is associated with an estimated 49.58-unit increase in diabetes progression.

BP Coefficient (0.90, $p < 0.001$): Holding BMI and S5 constant, each 1 mm Hg increase in mean arterial blood pressure is associated with an estimated 0.90-unit increase in diabetes progression.

Part C: Model Comparison and Critical Evaluation

Compare the simple and multiple models:

Explanatory Power: The multiple model substantially improves explanatory power compared to the simple model. R^2 increases from 0.3439 to 0.4801, and adjusted R^2 increases from 0.3424 to 0.4765. This confirms that adding blood pressure and serum lipid S5 accounts for an additional 13.6% of the variance in disease progression beyond BMI alone.

Model Precision: The residual standard error (sigma) decreases from 62.52 in the simple model to 55.78 in the multiple model, showing that predictions from the three-predictor model are closer to actual patient outcomes.

Critical Assessment (Is higher R-squared automatically better?):

A model with a larger R^2 is not automatically superior in every respect. Adding more predictors will almost always increase R^2 purely by chance, which can lead to overfitting, excessive model complexity, and unnecessary clinical testing costs. However, in this case, the increase in adjusted R^2 is substantial, all three predictors remain highly statistically significant ($p < 0.001$), and each represents a distinct, clinically meaningful biological mechanism (obesity, vascular pressure, and dyslipidemia). Therefore, the multiple model is genuinely superior both statistically and clinically.

Question 5: Evaluate the Analysis

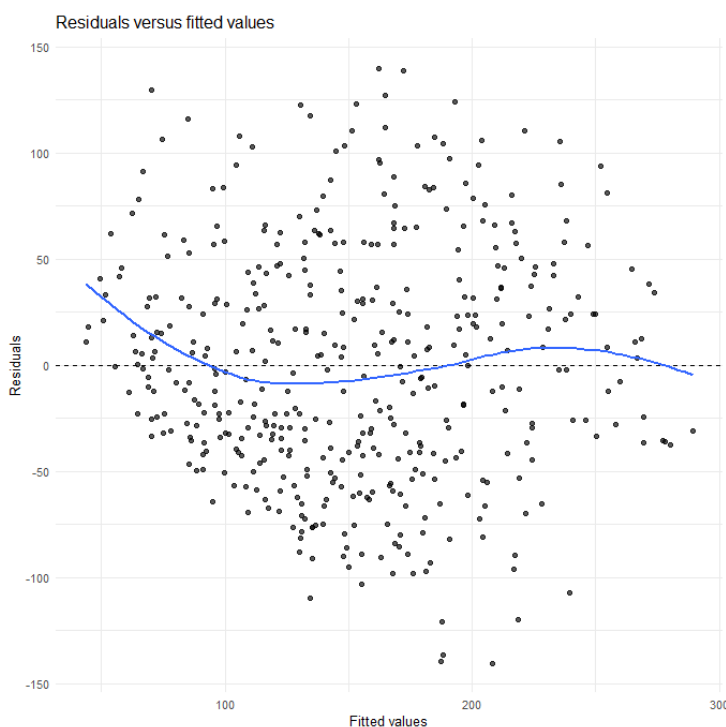


Figure 7: Residuals versus Fitted Values

with no severe funneling to indicate heteroscedasticity. However, a slight upward dispersion is noticeable at higher fitted progression levels, and the lower bound of progression scores creates a slight truncation effect at very low fitted values. Overall, the plot does not reveal severe violations that would invalidate ordinary least squares estimation.

Multicollinearity Assessment (VIF and Correlation)

Multicollinearity was evaluated using Variance Inflation Factors (VIF) alongside the pairwise correlation matrix:

VIF Estimates: The calculated VIF values for the three predictors are modest: 1.34 for BMI, 1.27 for blood pressure (BP), and 1.34 for serum marker S5.

To ensure that the multiple regression model (Model 2) provides trustworthy inferences, diagnostic evaluations were conducted examining residual behavior, potential multicollinearity, statistical uncertainty, and practical limitations.

Residual Diagnostics and Model Assumptions

A residuals-versus-fitted values diagnostic plot was constructed to assess linearity and homoscedasticity (Figure 7: Residuals versus Fitted Values):

Pattern and Linearity: The residuals are distributed relatively evenly above and below the horizontal reference line at zero across the entire range of fitted values. The smoothing trend line remains close to zero without displaying severe curvature, indicating that the linear specification is reasonable for this predictor combination.

Spread and Homoscedasticity: The vertical spread of the residuals appears broadly stable across fitted values,

Interpretation: All VIF values remain well below common diagnostic thresholds (such as 5 or 10), and even well below conservative limits of 2.5. Although BMI, BP, and S5 exhibit moderate pairwise correlations with one another (correlations ranging between 0.24 and 0.39), these correlations do not create problematic variance inflation. Consequently, the standard errors of the regression coefficients are not unduly inflated, and the unique statistical contribution of each clinical variable can be estimated with confidence.

Statistical Uncertainty and Confidence Intervals

To account for sampling variability, 95% confidence intervals were evaluated for the multiple regression estimates:

BMI Coefficient: The estimated slope is 6.50 with a 95% confidence interval of [5.13, 7.87] ($t = 9.32$, $p < 0.001$). Because the entire confidence interval lies comfortably above zero, we can conclude with high statistical confidence that higher BMI is associated with higher diabetes progression, with the true population increase falling between approximately 5.1 and 7.9 progression points per unit increase in BMI when holding BP and S5 constant.

S5 (Serum Lipid Marker) Coefficient: The estimated slope is 49.58 with a 95% confidence interval of [38.00, 61.15] ($t = 8.42$, $p < 0.001$). The lower bound indicates that even in the most conservative plausible scenario, an increase of 1 unit in log serum triglycerides is associated with at least a 38-point rise in disease progression.

BP (Blood Pressure) Coefficient: The estimated slope is 0.90 with a 95% confidence interval of [0.48, 1.33] ($t = 4.17$, $p < 0.001$), indicating a statistically robust positive association.

Statistical Significance versus Practical Clinical Importance

While all three predictors are highly statistically significant ($p < 0.001$), statistical significance must be distinguished from clinical relevance:

Magnitude of Effect: S5 and BMI show large, clinically meaningful estimated effects on progression. A moderate increase in BMI (for example, 4 kg/m²) is associated with an expected increase of approximately 26 progression points, which represents about one third of the standard deviation of disease progression in this cohort.

Practical Interpretation of BP: Blood pressure has a small per-unit slope (0.90 points per mm Hg). While statistically reliable, a 5 mm Hg shift in blood pressure yields a relatively modest predicted shift of 4.5 points on a 25-to-346 scale. Thus, statistical certainty does not always equate to large predictive influence in isolation.

Model Limitations

Several fundamental analytical and observational limitations must be noted:

Unexplained Variation: The multiple model explains approximately 48% of the variance ($R^2 = 0.4801$), leaving more than half (52%) of the variation in disease progression unexplained. Key unmeasured biological and lifestyle factors, such as diet, physical activity, genetic markers, medication adherence, and diabetes duration, are absent from the dataset.

Correlation vs. Causation: The findings reflect observational associations rather than causal mechanisms. Reducing BMI or blood pressure cannot be assumed to reduce progression by the exact estimated quantities without experimental intervention trials.

Residual Standard Error: The model's residual standard error is 55.78 points. Given that the overall standard deviation of progression is 77.09, individual patient predictions carry considerable uncertainty and should not be used as definitive individual clinical prognostications.

Question 6: Draw Appropriate Conclusions and Reflect on Limitations

The final objective of this study is to synthesize the statistical findings into sound clinical conclusions while rigorously evaluating the limitations of the data and modeling framework. Responsible interpretation requires distinguishing between claims that are strictly supported by empirical evidence and claims that overreach the analytical design.

- **Is this dataset experimental or observational?**

This dataset is strictly observational. The data consist of baseline clinical characteristics and lab measurements recorded from 442 diabetes patients without any experimental manipulation, randomized treatment allocation, or controlled clinical intervention.

- **Can the regression coefficients be interpreted causally?**

No, the regression coefficients cannot be interpreted causally. The estimated slopes represent observational associations rather than direct causal effects. For example, while the multiple regression model shows that each 1-unit increase in BMI is associated with an average increase of 6.50 progression points holding blood pressure and S5 constant, this does not imply that an intervention reducing a patient's BMI will automatically cause an equivalent reduction in disease progression. Observational data cannot rule out reverse causation or unmeasured confounding.

- **Could there be confounding variables that are not included in the dataset?**

Yes, substantial confounding is likely. Crucial lifestyle, behavioral, and clinical variables are omitted from the dataset, including patient dietary habits, physical activity, alcohol consumption, smoking status, socioeconomic background, duration of diabetes, and current medication use (such as metformin, insulin, or antihypertensives). If any of these unmeasured factors correlate simultaneously with both the predictors (e.g., BMI or blood pressure) and disease progression, the estimated regression coefficients may suffer from omitted variable bias.

- **Could measurement issues affect the results?**

Yes. First, baseline predictors such as blood pressure and serum lipid levels are subject to biological variation and measurement error from single-occasion clinical visits. Second, the response variable is a quantitative composite measure of progression one year after baseline, which does not directly capture clinical hard endpoints such as cardiovascular events, hospitalizations, or organ damage (nephropathy, retinopathy). Any unreliability or imprecision in recording these values introduces noise into the model.

- **Would the results necessarily generalise to another population?**

Not necessarily. The sample is limited to 442 individuals from a specific, unspecified cohort without demographic details regarding ethnicity, nationality, or access to care. Differences in genetics, healthcare quality, baseline diets, and treatment protocols mean these empirical relationships may not generalize reliably to other populations, older or younger age groups, or patients at different stages of diabetes without external replication.

- **Which conclusions are supported by the evidence?**

1. Body mass index (BMI), blood pressure (BP), and serum marker S5 each display statistically significant, positive linear relationships with one-year diabetes progression in this patient cohort (all $p < 0.001$).
2. Combining these three clinical measures into a multiple regression model noticeably improves predictive power over a single-predictor model, raising explained variance from 34.39% to 48.01% and reducing residual standard error from 62.52 to 55.78 points.
3. Multicollinearity among BMI, BP, and S5 is low (all VIF values are around 1.3), meaning each variable contributes distinct explanatory value within this model.

- **Which claims would go beyond the evidence?**

1. Claiming that lowering a patient's BMI or blood pressure will directly prevent or reverse diabetes progression (confusing correlation with causation).
2. Using this regression model as an exact prognosis tool for individual patients, given that 52% of the variance remains unexplained and the residual standard error is relatively large (55.78 points).
3. Recommending broad clinical treatment policies or universal clinical thresholds based solely on this limited, observational 442-patient sample.

